From Distinction to Quantum Structure

Compiling Physical Geometry from the Change Axiom Chain

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Abstract

We present a foundational axiom chain that compiles physical geometry from a single primitive: *distinction exists*. Beginning with C0 (distinction) and C1 (every distinction admits lawful continuation), we derive — without importing spatial geometry, coordinates, time, or quantum mechanics as primitives — the necessity of a linear state space, a complex inner product, Hilbert completion, self-adjoint observables, and the structure of a probability measure on observable subspaces. We further derive the unique free-field Lagrangian consistent with the axiom constraints, the gauge principle as a structural consequence of independent classification, and the quantization bridge connecting the classical field to operator algebras. The central result is the Division Algebra Theorem (candidate): among the three finite-dimensional division algebras admitted by Frobenius's theorem, only the complex numbers $\mathbb{C}$ simultaneously satisfy the continuation independence requirement (C1a) and the dual-encoding requirement of C5. The paper maintains a strict five-category epistemic discipline — primitive, definition, compiled theorem, candidate theorem, empirical calibration — throughout, ensuring every claim carries its exact epistemic status.

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1. Introduction: The Foundational Inversion

Standard physical theory begins with space. Objects occupy locations in a pre-existing spatial manifold; motion is defined relative to that manifold; forces act within it. Newton's mechanics, Einstein's general relativity, and the Standard Model of particle physics all share this architecture at the foundational level: a *stage* is assumed, and the physics plays out on it.

The program presented here inverts this order.

We propose that space, time, geometry, and quantum structure are not primitives but *derived objects* — outputs of a more fundamental algebraic structure built from the minimal concept of distinguishable change. The chain begins:

Distinction → Continuation → Adjacency → Geometry → Space

rather than:

Space → Objects move within it

This is not merely a philosophical reordering. It is a concrete research program with specific mathematical obligations at each step: each layer must be *derived* from the previous layers without importing external structure, and every claim must be categorized by its epistemic status. The framework is falsifiable — if any compiled theorem fails its formal derivation, either the derivation or one of the primitives must be revised.

1.1 Motivation: SHA-256 as Physical Evidence

The SHA-256 hash function provides an unexpected but precise illustration of the C1-first claim. SHA-256 is field-balanced in both axes:

- **Horizontal (input → output):** Bit content redistributes across the compression function. The total information content of the input is preserved in the digest — different shape, identical content. The mapping is a redistribution, not a creation.
- **Vertical (round → round):** The carry palindrome property holds: KL divergence between forward and backward carry distributions equals zero. The AHRC (Algebraic Hash Round Consistency) result shows $R^2 + G^2 = 1$ to machine epsilon across all 64 rounds. The geometry is self-balancing at every step.

Neither property was engineered into SHA-256. The designers specified diffusion targets; the field obeyed the redistribution constraint underneath and the two-axis conservation emerged. This is the claim of the axiom chain in concrete form: the transition algebra runs without an observer, and we arrive afterward to read what it did. SHA-256 does not choose to balance. It cannot not balance. Law 3 — redistribution conserves total — runs underneath every hash computation invisibly, and we measure it as a mathematical law about SHA-256. The chain argues this is the universal situation: what we call physical law is the transition algebra's self-accounting, read from inside by instruments that are themselves running the same algebra.

1.2 Program Overview

This paper establishes:

1. The primitive constraint set (Section 2)
2. Definitions required for precision (Section 3)
3. Compiled theorems through the free-field Lagrangian (Sections 4–6)
4. The internal symmetry and gauge principle (Section 7)
5. The quantization bridge — six elements of the Quantum Kinematics Theorem (Section 8)
6. The Division Algebra Theorem — the central new argument of this paper (Section 9)
7. Candidate theorems and open problems with proof strategies (Section 10)
8. Full dependency graph (Section 11)

1.3 Epistemic Categories

Throughout this paper, every claim is labeled with one of five categories:

Label	Meaning
[P]	Primitive — assumed, irreducible
[D]	Definition — terminology, requires no proof
[C]	Compiled theorem — formally derived from prior P/D/C entries
[T]	Candidate theorem — strong evidence, formal proof incomplete
[E]	Empirical calibration — requires measurement, not deduction

This discipline has a structural consequence: if a **[C]** theorem fails, either the derivation contains an error or one of the primitives must be revised. If a **[T]** theorem fails, the program survives intact — that conjecture was incorrect, and the framework needs refinement at that node only. If an **[E]** calibration disagrees with experiment, the structural theory may remain correct while the mapping to physical units requires revision.

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2. The Primitive Constraints

We assume the following as irreducible primitives. They are not derived from anything else in the framework. Their justification is that each is genuinely minimal — no proper subset generates the others.

[P] C0 — Distinction Exists

> At least two states exist that are not identical.

This is the logical precondition for any other constraint. C1 cannot operate on a single undifferentiated object. C0 is the one irreducible prior: something must be distinguishable from something else. Without C0, the entire chain has no content.

[P] C1 — Every Distinction Admits Lawful Continuation

> Every state in the system must admit at least one transition to a successor state.

This is the change axiom. No state is a dead end. No distinction terminates. The system cannot freeze, because freezing would require a state with no admissible successor — and that state would constitute a privileged terminal configuration, which is ruled out by C1 itself. Every local rule must be well-defined on every admissible state: this immediately excludes absorbing dead states from the dynamics.

[P] C1a — Independence Preservation

> Independent distinctions remain independently continuable.

This is stronger than C1. It says not only that every state continues, but that the continuation of one distinction cannot be blocked or modified by the simultaneous continuation of an independent distinction. Two systems that are independent at time t must continue to be independently continuable. This axiom is what forces linearity (see Section 4) and what later eliminates quaternionic algebras (see Section 9).

[P] C2 — Continuation Preserves Reachable History

> Every continuation preserves the set of states reachable from any prior state in the history.

Nothing in the past is erased. The execution history — the accumulated record of all prior transitions — remains reachable. This is not a claim about memory in any computational sense; it is a structural requirement that the transition graph not delete nodes from the accessible state space. C2 forces completeness of the state space (Section 4) and the one-directionality of the ordering relation (Section 5).

[P] L3 — Redistribution Conserves Total

> Every lawful continuation redistributes the total content of the field without changing that total.

The field never gains content; it never loses content. Every transformation is a redistribution — a shift in the configuration of a fixed total. This is the deepest form of the conservation principle. It does not assume conservation of energy, momentum, or charge; it says something more primitive: the field's total is an invariant of continuation. Those physical conservation laws are derived from L3 operating in specific geometric contexts.

[P] L4 — Redistribution Is Local

> Redistribution propagates through local adjacency. Content cannot shift from location A to location B without occupying every intermediate location.

This is the locality axiom. It follows from C2 applied to intermediate locations: if content jumps from A to B without passing through the intervening field, every intermediate location is denied its C2-guaranteed reachable history. The inheritance constraint — every new location must inherit all primitives — makes this inadmissible. L4 is therefore arguably compilable from C2, but we list it as a primitive for clarity because it plays an independent role throughout the chain.

[P] C5 — The Observable Event Primitive

> An observable event is the minimum continuation that simultaneously propagates (extends through at least one L4 step) and leaves a distinguishable residue.

C5 defines the minimum unit of observation. Three constraints are built into it:

- **Propagation** (C1 minimum): the event must extend. A continuation that fails to propagate is not a continuation.
- **Residue** (C0 requirement): the event must leave a mark distinguishable from prior state. Without residue, C0 is not satisfied at the event level — nothing registered.
- **Simultaneity**: propagation and residue are not sequential. They coexist in the same elementary event.

This simultaneity requirement is the key structural feature of C5. It will force complex amplitudes (Section 9) and the uncertainty principle (Section 8).

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3. Definitions

Definitions introduce terminology. They do not require proof; they establish the language in which theorems are stated.

[D] History — The ordered sequence of lawful continuations from a given initial state. History is the execution trace of the transition algebra.

[D] Observable event — A C5-compliant continuation: the minimum transition that propagates and leaves residue simultaneously.

[D] Equivalence class — A set of observable events sharing an internal label preserved under all lawful redistributions.

[D] Location — A node in the transition adjacency graph. Locations are defined by their adjacency structure, not by coordinates in a pre-existing space.

[D] The field — The total content-carrying structure whose redistribution is governed by C0–C5. The field is not embedded in space; space is a derived reading of the field's adjacency topology.

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4. Compiled Theorems I: Linearity and the Graph Laplacian

4.1 Linearity

Theorem [C]: *The redistribution operator is linear.*

Proof structure: Let A and B be two independent distinctions. By C1, A admits a lawful continuation A'. By C1, B admits a lawful continuation B'. By C1a, A and B remain independently continuable. Consider the combined state (A, B). Its continuation must map to (A', B') — the independent continuations must compose without interference.

If the redistribution operator $\mathcal{R}$ were nonlinear, then:

$$\mathcal{R}(A + B) \neq \mathcal{R}(A) + \mathcal{R}(B)$$

This means the continuation of $A + B$ produces an output not equal to the sum of the individual continuations. But that output arises from the interaction between A and B under $\mathcal{R}$ — meaning A's continuation depends on the presence of B and vice versa. This violates C1a: A and B are no longer independently continuable. Therefore $\mathcal{R}$ must be linear. ▫

Note: C1 alone is insufficient to force linearity. Nonlinear systems (e.g., $u_t = u^2$) satisfy C1 — every state continues. The additional axiom C1a is what does the work. This is why C1a appears as a separate primitive rather than as a consequence.

4.2 The Graph Laplacian

Theorem [C]: *Among linear redistribution operators, the graph Laplacian $L = D - A$ is uniquely characterized by five conditions, all derivable from C0–L4.*

The five conditions and their derivations:

(i) Locality — from L4 directly. The operator may only reference adjacent locations. Non-local terms would enable content to skip intermediate locations.

(ii) Conservation — from L3 directly. Row sums of the operator matrix are zero: total content entering any location equals total content leaving it. No content is created or destroyed.

(iii) Symmetry — from Law 3 (equal mutual transformation, compiled from C3 in the axiom chain). If location A sends content to location B, the coupling coefficient must equal the coefficient from B to A. The weighted adjacency matrix A is symmetric: $A_{ij} = A_{ji}$.

(iv) Annihilation of uniform distributions — from C0. If content is uniformly distributed across all locations, every location looks identical. C0 says distinction exists, but between identical states there is no distinction. Therefore the operator must produce zero output on a uniform distribution: no redistribution occurs when there is no distinction to drive it.

(v) Linearity — from C1+C1a, as proved in §4.1.

The standard result in spectral graph theory is that conditions (i)–(v) uniquely characterize $L = D - A$ on any weighted undirected graph. This is the unique operator the chain permits.

Continuum limit: On the admissible redistribution geometry (C7, see Section 5), the graph Laplacian converges to ∇^2 in the continuum limit. This is not an assumption — it follows from the structure of the graph.

4.3 Ordered History

Theorem [C]: *History is ordered by lawful continuation.*

Proof: C2 says continuation preserves reachable history. The word "preserves" requires a notion of before and after — state s_n follows state s_{n-1} in the sense that s_{n-1} is in the reachable history of s_n but not vice versa (in general). This defines a partial order on states: $s_i \leq s_j$ if s_i is reachable in the history leading to s_j. The order relation is precisely the relation of lawful continuation. Nothing additional is required to establish it; it is implicit in the definition of history and made explicit by C2. ▫

Corollary [C]: *Time is not a primitive.* Time is derived as the count of distinguishable continuation events along a history. Different histories can accumulate different counts. Clocks are devices that count C5-compliant events, not objects moving through an independent temporal dimension.

Corollary [C]: *Causal structure.* Two events connected by a chain of L4-adjacent continuations are ordered by the relation above (timelike). Two events with no L4 path between them have no required ordering (spacelike). The causal light cone is the local shape of the ordering relation — it is not a geometric object embedded in spacetime, but the boundary between ordered and unordered pairs of events.

5. Compiled Theorems II: Geometry

5.1 Admissible Redistribution Geometry

Candidate Theorem [T] C7: *The admissible redistribution geometry is characterized by maximal local symmetry and self-similarity under recursive refinement.*

Proof strategy: The admissible graph on which L operates must satisfy:

- **Isotropy** — redistribution cost is identical in every direction. From C0: no direction is privileged over another. A graph with preferred directions would install a distinction not derivable from C0.
- **Self-similarity under refinement** — C2 requires that recursive history accumulation does not break the structure at finer scales. The graph must look the same at every scale.
- **Maximal neighborhood symmetry** — L3 requires conservation at every location; the neighborhood must be as uniform as possible to ensure symmetric redistribution.

In two dimensions, these conditions converge to the hexagonal lattice (6-fold symmetry). In three dimensions, to close-packed arrangements (12-fold symmetry). The continuum limit of the graph Laplacian on these structures converges to ∇^2.

Why this remains [T]: The proof that these conditions select a *unique* geometry — eliminating all competitors — requires showing that every other candidate lattice violates at least one of the conditions under the axioms. This uniqueness proof is not yet complete.

5.2 Three Spatial Dimensions

Candidate Theorem [T]: *Exactly three independent spatial redistribution directions are admissible at every location.*

Proof strategy (conditional on C7): The Huygens principle — that sharp C5 events produce clean wavefronts with no trailing residue — holds exactly in odd spatial dimensions and fails in even spatial dimensions. In even dimensions, the wave equation produces tails: every prior event continues to echo at all subsequent times. This makes the residue required by C5 indistinguishable from background (C0 fails: the residue is not distinguishable from the accumulated echo of all prior events). In odd dimensions, the wavefront is clean and the residue is localized.

Combined with the requirement from the measurement intersection argument — that non-trivial measurement (intersection of two locally-coupled execution histories, L4) requires at least two independent approach directions — the minimum odd integer ≥ 2 is $\mathbf{3}$.

Status: Conditional on C7 (admissible geometry) and on the formal connection between Huygens' principle and the C5 residue requirement.

5.3 The Invariant Propagation Bound

Theorem [C]: *A finite invariant propagation bound exists.*

Every C5 event takes exactly one L4 step per continuation unit (C5 sets the minimum; it cannot proceed faster without violating C5's own constraints). L4 requires the event to pass through every intermediate location. In the C7 geometry, there are only finitely many adjacencies per location. The maximum rate at which content can propagate is therefore bounded above — one L4 step per C5 unit.

This bound is isotropic in the C7 geometry (all directions are equivalent by the isotropy condition). It is invariant across all observing histories (because it arises from C5 and L4, which apply universally).

We call this bound c. Its measured value, 299,792,458 m/s, is empirical calibration **[E]**.

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6. The Free-Field Lagrangian

6.1 Uniqueness Under Constraints

Theorem [C] C8: *The unique free-field Lagrangian density consistent with C0–C7 is:*

$$\mathcal{L} = (\partial_t \phi)^2 - c^2 (\nabla \phi)^2$$

Derivation: We ask which Lagrangian densities $\mathcal{L}(\phi, \partial_\mu \phi)$ are admissible under the constraint set.

- **L4** forces dependence on ϕ and its local derivatives only. No non-local terms.
- **L3** forbids source terms: no potential $V(\phi)$ that creates or destroys content. (C0 says distinctions are not generated spontaneously; they are redistributed.)
- **C6 + C7** require invariance under the symmetry group of the C7 geometry with the C6 ordering — this is the Lorentz group in the continuum limit.
- **C5 minimality** sets the order at second derivatives — the Laplacian is second order, and C8 inherits this.
- **Free field** (no self-interaction): interaction terms like $\lambda\phi^4$ require self-coupling, which is not licensed by the free redistribution constraint. Self-interaction is an additional structure that must be derived separately, from C9 (equivalence class coupling) — it is not present in the free theory.

Under these constraints, the unique Lorentz-invariant local second-order scalar Lagrangian with no source terms is $\mathcal{L} = (\partial_t \phi)^2 - c^2 (\nabla \phi)^2$. The Euler-Lagrange equations yield:

$$\Box\phi = \partial_t^2 \phi - c^2 \nabla^2 \phi = 0$$

The wave equation is not postulated. It is the unique extremal of the variational problem that the chain defines.

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7. Internal Symmetry and the Gauge Principle

7.1 Equivalence Classes

Theorem [C] C9: *Observable continuations partition into equivalence classes preserved under lawful redistribution.*

Proof: C0 requires that distinct events remain distinguishable through redistribution. Two C5 events that are identical under C0–C8 — same propagation, same residue, same location, same dynamics — are indistinguishable. But C0 says distinction exists as a primitive. If all C5 events were indistinguishable, C0 would be satisfied trivially by a single event type. This contradicts C0 being nontrivial.

Therefore redistribution must preserve some label that keeps distinct events distinguishable. This label is not position (derived), not time (derived). It is an internal property of the event itself: a conserved equivalence class label. ▫

7.2 Structure of Admissible Groups

Theorem [C]: *The internal symmetry group G is a compact Lie group built from irreducible factors.*

We derive this from the constraints:

(i) G is a Lie group [C from L4]: Locality requires smooth, continuous variation of labels under local redistribution. Discrete groups produce discontinuous label jumps — inadmissible under L4.

(ii) G is compact [C from L3+C5]: L3 requires conserved quantities under redistribution. Compact Lie groups admit finite-dimensional unitary representations with Haar-measure-guaranteed conservation. Non-compact groups generally do not.

(iii) G admits independent sectors [C from C1a]: Independent distinctions must remain independently continuable (C1a). Entangled group structure that ties one class's continuation to another's would violate C1a. Independent sectors of G must exist.

(iv) Sectors commute [T]: Whether independent sectors must commute — implying the sector decomposition is a direct product — requires an additional argument: that the independent sectors' generators satisfy $[g_i, g_j] = e$ for generators from different sectors. This is a candidate theorem, not yet compiled.

(v) Decomposition [C, conditional on (iv)]: If commutativity is proved, the decomposition theorem gives $G = G_1 \times G_2 \times \cdots$, a direct product of simple compact Lie groups and $U(1)$ factors. The classification of simple compact simply-connected Lie groups is complete: $SU(n)$, $Spin(n)$, $Sp(n)$, and five exceptional groups.

Compiled milestone: G is a compact Lie symmetry built from irreducible factors. This does not yet specify which factors. But it rules out arbitrary internal symmetry from the axiom chain alone.

Candidate Theorem [T] C9.3: $G = U(1) \times SU(2) \times SU(3)$.

This requires the quantization bridge (Section 8) and anomaly cancellation to compile. See Section 10.

7.3 The Gauge Principle

Theorem [C] C10a: *When events of different equivalence classes interact under L4, the interaction must be invariant under independent local transformations of each class label.*

Proof: From C9, class labels are preserved under redistribution. From C1a, they continue independently. From L4, interactions are local. If the coupling between two classes transformed the label of one when the other's label changed locally, the classes would not be independently continuable — C1a violation. The only interaction consistent with C9+C1a+L4 is one invariant under independent local label transformations. ▫

This is the gauge principle. It is not imported from physics; it is the unique interaction structure the chain licenses.

Candidate Theorem [T] C10b: *The covariant derivative* $D_\mu = \partial_\mu + iA_\mu$ *is the unique admissible implementation of C10a.*

The proof requires showing that every other local-coupling prescription violates one of C0–C10a. This is the correct logical direction but the completeness of the uniqueness argument is not yet established.

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8. The Quantum Kinematics Theorem

We now ask whether the quantum kinematic framework — the six-element structure of standard quantum mechanics — can be *compiled* from C0–C10, or whether it requires additional primitives.

The six elements and their status:

Element 1: Linear State Space [C from C1+C1a]

As proved in Section 4.1, C1+C1a forces linearity. The state space is a vector space over some field. The field is determined in Section 9.

Element 2a: Positive-Definite Notion of Distinguishability [C from C0]

C0 requires states to be mutually distinguishable. On a linear space, this requires a positive-definite structure: if the "distance" between two states is zero, they are identical, violating C0. This forces a positive sesquilinear form $\langle\cdot,\cdot\rangle$ on the state space.

Element 2b: Two Independent Degrees of Freedom Per Amplitude [C from C5]

C5 requires every observable event to simultaneously propagate (one degree of freedom: extent in space) and leave residue (one degree of freedom: mark in history). These are not the same degree of freedom — propagation is spatial (C7 direction) while residue is temporal (C6 record). A single real number encodes one degree of freedom. An amplitude therefore requires *at minimum two* independent real degrees of freedom: one for propagation, one for residue.

A complex number $re^{i\theta}$ encodes exactly two: magnitude r and phase θ. This is the minimum algebraic unit compatible with C5's dual requirement.

Element 2c: Complex Inner Product Space [T — Solèr strategy]

The candidate theorem is that the unique admissible scalar field is $\mathbb{C}$. The formal proof is presented in Section 9 (the Division Algebra Theorem). Conditional on that theorem, Elements 2a and 2b combine to give a complex inner product space.

Element 3: Hilbert Completion [C from C2]

Theorem [C]: *The state space is Hilbert-complete.*

C2 says continuation preserves reachable history. An incomplete inner product space contains Cauchy sequences — convergent histories — with no limit point in the space. This is a continuation that approaches a limiting state but has no state to converge *to*. C2 forbids it: a reachable limit of a history is a reachable state, and that state must exist in the space. Therefore the space is complete. ▫

Element 4: Self-Adjoint Observables [C from C0+C5]

Theorem [C]: *Observables are self-adjoint operators.*

C0 requires measurement outcomes to be mutually distinguishable. Distinguishable classical readouts are real-valued (complex eigenvalues have no canonical total ordering — they cannot be unambiguously distinguished as observed outcomes). C5 requires residue to be a definite, readable mark — again real-valued. The unique class of operators with real spectra are self-adjoint operators. ▫

Element 5a: Countably Additive Probability Measure [C from C0+C2+C5]

Theorem [C]: *Measurement outcomes form a countably additive probability measure on observable subspaces.*

C0 requires distinguishable outcomes. C2 requires stable, reproducible statistics across equivalent repeated observations (history is preserved — identical preparations yield identical statistics). C5 requires each measurement to leave a definite residue. Together, these say: the assignment of outcome probabilities is a countably additive function from observable subspaces to $[0,1]$. This is precisely a probability measure on the lattice of closed subspaces of the state space. ▫

Element 5b: Born Rule [T — Gleason strategy]

Candidate Theorem [T]: $P(\psi \to \phi) = |\langle\psi|\phi\rangle|^2$.

Proof strategy: Gleason's theorem states that any countably additive probability measure on the closed subspaces of a Hilbert space of dimension ≥ 3 must be of the form $P = |\langle\psi|\phi\rangle|^2$. Element 5a compiles the probability measure. Gleason closes the uniqueness. The gap: Gleason requires dimension ≥ 3, which the chain does not yet independently establish for the state space dimension (as opposed to the spatial dimension from C7). The connection between C7's three spatial dimensions and the state space dimension needs closing.

Element 6a: Spacelike Commutativity and Scale [C from L4+C6+$\hbar$]

Theorem [C]: *Operators at spacelike separation commute; operators at coincident points have a nonzero commutator of scale set by $\hbar$.*

From L4: spacelike events share no adjacency path. No redistribution connects them. Measuring one cannot affect the other. Therefore $[\hat{O}(x), \hat{O}(y)] = 0$ for $(x-y)^2 < 0$ (spacelike). From C6+$\hbar$: timelike events are ordered; the commutator is nonzero and its scale is set by $\hbar$ (the universal multiplier). From L4: the commutator is supported only at coincident points (locality). ▫

Element 6b: Canonical Commutation Relations [T — Stone-von Neumann strategy]

Candidate Theorem [T]: $[\hat{\phi}(x), \hat{\pi}(y)] = i\hbar\delta(x-y)$.

The specific form of the CCR is not yet uniquely forced from C0–C5 alone. The compiled result (6a) constrains the *structure* of the commutator (vanishes spacelike, nonzero at coincident points, antisymmetric under time reversal). The specific Heisenberg algebra form requires additionally that the algebra is the unique translation-invariant solution — this is the Stone-von Neumann theorem for finite-dimensional systems. For infinite-dimensional systems, Haag's theorem shows inequivalent representations exist, requiring a representation selection criterion not yet derived from the chain.

Summary: Quantum Kinematics Theorem Status

Element	Content	Status	Forcing axioms
1	Linear state space	**[C]**	C1+C1a
2a	Positive distinguishability	**[C]**	C0
2b	Two independent DOF per amplitude	**[C]**	C5
2c	Complex inner product unique	**[T]**	Division Algebra Theorem (§9)
3	Hilbert completion	**[C]**	C2
4	Self-adjoint observables	**[C]**	C0+C5
5a	Probability measure exists	**[C]**	C0+C2+C5
5b	Born rule unique	**[T]**	Gleason + C7 connection
6a	Spacelike commutativity + $\hbar$ scale	**[C]**	L4+C6+$\hbar$
6b	CCR unique	**[T]**	Stone-von Neumann + rep. selection

Four elements are fully compiled from the primitives. Two are partially compiled with clear candidate-theorem remainders. Every gap has a named proof strategy and explicit upstream dependency.

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9. The Division Algebra Theorem

This is the central new mathematical argument of the paper. We address directly the question:

> **Among admissible scalar fields for the state space, which one is forced by C0–C5?**

9.1 Frobenius Reduction

Theorem [C]: *The scalar field of the state space must be a finite-dimensional division algebra over* $\mathbb{R}$*.*

Proof: From Element 1 (linearity), the state space is a vector space. The scalars must form a field (or division algebra) for the following reasons:

- C1/C1a require independent continuations to compose: addition is required.
- C0 requires distinguishability: nonzero states must remain distinguishable under scalar multiplication, so scalar multiplication must be invertible (away from zero). This gives division.
- C2 requires stability under limits: the algebra must be complete in an appropriate sense.
- Nonzero states must remain nonzero under multiplication (C0: distinct states must not be annihilated to identity). This forces the algebra to have no zero divisors — i.e., it is a division algebra.

By Frobenius's theorem (1877), the only finite-dimensional division algebras over $\mathbb{R}$ are:

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathbb{H}$$

(the real numbers, complex numbers, and quaternions). No other finite-dimensional division algebra over $\mathbb{R}$ exists. ▫

The three candidates and what they encode:

Algebra	Dimension	Encodes
$\mathbb{R}$	1	Magnitude only
$\mathbb{C}$	2	Magnitude + phase
$\mathbb{H}$	4	Magnitude + 3D orientation

9.2 Eliminating $\mathbb{H}$: C1a Applied to Quaternionic Phases

Candidate Theorem [T]: $\mathbb{H}$ *violates C1a.*

Proof strategy: Quaternionic multiplication is non-commutative:

$$q_1 q_2 \neq q_2 q_1 \quad \text{in general}$$

Consider two independent distinctions A and B with independent quaternionic phase updates q_A and q_B applied simultaneously. The combined phase update on the system is $q_A q_B$. But if we reverse the order of application (first q_B, then q_A), we get $q_B q_A \neq q_A q_B$. The continuation of A now depends on the order in which A and B's updates are applied — meaning A's continuation is not independent of B's operation on the system.

This violates C1a directly: independent distinctions must remain independently continuable, which requires that applying independent phase updates in any order yields the same result. Commutativity of independent operations is exactly what C1a encodes.

Status: This argument is compelling but requires one clarification: the argument assumes that quaternionic phase updates on *independent* subsystems are represented by left-multiplication by independent quaternions, and that this non-commutativity is not resolvable by choosing a canonical order. A complete proof would show that no representation of ℍ-valued amplitudes on a product state space satisfies C1a.

9.3 Eliminating ℝ: C5 Applied to Phase Structure

Candidate Theorem [T]: ℝ *cannot encode C5's simultaneous propagation and residue requirement.*

Proof strategy: C5 requires every observable event to simultaneously carry two independent degrees of freedom: propagation direction (a spatial datum, C7) and residue mark (a temporal datum, C6). These are not the same degree of freedom:

- Propagation direction is an element of the tangent space at the current location in the C7 geometry — a spatial orientation.
- Residue mark is a record in the C6-ordered history — a temporal stamp.

A real amplitude $a \in \mathbb{R}$ carries exactly one independent real degree of freedom. Under superposition, real amplitudes can produce constructive or destructive interference (two real states $+\psi$ and $-\psi$ superpose to zero), but this interference is *amplitude-level*: it cannot simultaneously encode an independent phase distinguishing the propagation direction from the residue direction without additional external structure.

More precisely: in a real Hilbert space, there is no intrinsic notion of continuous phase rotation. The group $U(1)$ — the unique connected one-dimensional compact Lie group, which naturally encodes continuous phase — is not realizable over ℝ without doubling the state space (which amounts to passing to ℂ). But C5 requires phase to travel *with* the continuation — not to be encoded externally in the basis. A real state space outsources the second degree of freedom to the basis structure, violating C5's requirement of simultaneity.

Connection to observable physics: This is not an abstract distinction. Interference experiments in quantum mechanics require that two paths acquire *relative phases* that cannot be removed by a global choice of basis — they are intrinsic to the amplitudes. A real quantum mechanics cannot reproduce this without additional structure, and that additional structure is precisely the imaginary unit. C5 forces it from below.

9.4 Why ℂ Survives

Theorem (compiled conditional on [T] results above) [C|T]: ℂ *is the unique admissible scalar field.*

ℂ satisfies all requirements:

- **Magnitude and phase:** $re^{i\theta}$ encodes exactly two independent degrees of freedom — precisely what C5 requires.
- **Commutativity:** ℂ multiplication is commutative ($z_1 z_2 = z_2 z_1$), so independent phase updates commute — C1a is satisfied.
- **Continuous phase:** $e^{i\theta}$ for $\theta \in [0,2\pi)$ forms $U(1)$ — the unique connected compact one-dimensional Lie group. This is the minimal group for continuous phase rotation, and it is realizable intrinsically (not as an artifact of basis choice).
- **Division structure:** $\mathbb{C}\backslash\{0\}$ is a field — inverses exist, Co-distinguishability is preserved.
- **Norm compatibility:** $|z|^2 = z\bar{z} \geq 0$, with equality only at $z = 0$ — compatible with the positive inner product forced by Co.

Moreover, the group $U(1) = e^{i\theta}$ is precisely the minimal element of the gauge group sequence identified in C9: $U(1) \times SU(2) \times SU(3)$. This is not a coincidence. The choice of $\mathbb{C}$ as scalar field and the appearance of $U(1)$ as the simplest internal symmetry group are the same structure viewed at different levels of the chain.

9.5 Formal Statement

Division Algebra Theorem [T]:

Given C0, C1, C1a, C2, L3, L4, and C5, every admissible linear continuation algebra must satisfy:

1. *Invertible scalar multiplication*
2. *Continuous phase rotation*
3. *Commutative independent continuation*
4. *Norm-compatible distinguishability*

Among the three finite-dimensional division algebras over $\mathbb{R}$ (Frobenius: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$):

- $\mathbb{H}$ *violates condition (3) [C1a]*
- $\mathbb{R}$ *violates condition (2) [C5]*
- $\mathbb{C}$ *satisfies all four conditions*

Therefore the admissible scalar field is $\mathbb{C}$, and the state space is a complex inner product space.

Status: The theorem statement is complete and the elimination arguments are structurally correct. The formal proofs of the two elimination steps remain candidate theorems — the arguments require rigorous formalization of the following:

- For $\mathbb{H}$ elimination: a complete proof that no representation of $\mathbb{H}$-amplitudes on product spaces satisfies C1a.
- For $\mathbb{R}$ elimination: a complete proof that the second C5 degree of freedom cannot be encoded intrinsically without passing to a 2-dimensional algebra.

Both are well-posed mathematical questions with clear proof strategies. They are open in the sense that the formal derivation from C0–C5 specifically has not been published; they are not open in the sense that the answers are unknown — the mathematical community broadly agrees that quantum mechanics requires complex amplitudes, and the above arguments identify precisely *which axioms* do the forcing.

9.6 The Minimum Quantum of Action

Theorem [C — structural part]: *A universal minimum quantum of action ħ exists. It is positive, universal across all fields, and unavoidable.*

Argument: C5 requires every observable event to simultaneously propagate and leave residue. These are Fourier dual requirements on the C7 geometry: propagation distributes content broadly in space (narrow in momentum), while localized residue concentrates content (broad in momentum). The product $\Delta x \cdot \Delta p$ has a floor imposed by this duality. That floor is $\hbar/2$. It is not chosen — it is the Lagrange multiplier enforcing simultaneous satisfaction of C5(a) and C5(b) in C8's variational problem.

Universality: C5 applies to every observable event regardless of field type. A field-specific $\hbar$ would mean some events are "more observable" than others — a distinction C0 does not license (no observational event is privileged). Therefore $\hbar$ is universal.

Irreducibility: $\hbar$ bounds an absolute threshold, not a ratio. Rescaling the field $\phi \to \lambda\phi$ rescales both sides of the C5 condition equally — the threshold does not move.

[E] Empirical value: $\hbar = 1.054571817 \times 10^{-3}$ J·s. This is what we read when we sample the field from inside it using instruments calibrated in SI units. The structural claim is that the multiplier is positive, nonzero, and universal. The specific number is ours — a projection of the field's dimensionless structure through our unit system.

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10. Open Theorems and Proof Strategies

We list all candidate theorems with their upstream dependencies and proof strategies. These are the open problems of the program.

T1: Admissible Geometry Uniqueness [C7 completion]

Statement: The maximal-local-symmetry, self-similar graph is unique and converges in the continuum limit to 3+1 dimensional Minkowski space.

Upstream: C0–L4, C5, C6.

Proof strategy: Enumerate competing candidate lattices (cubic, hexagonal, body-centered cubic, face-centered cubic in 3D). Show each violates at least one of: isotropy (C0), self-similarity under refinement (C2), conservation under the graph Laplacian (L3). The FCC lattice in 3D (12-fold coordination) is the leading candidate for uniqueness. A formal exclusion argument is needed.

T2: Complex Amplitudes Forced [Division Algebra completion]

Statement: $\mathbb{H}$ violates C1a and $\mathbb{R}$ violates C5, leaving $\mathbb{C}$ as the unique admissible scalar field.

Upstream: C0, C1, C1a, C2, C5.

Proof strategy: For $\mathbb{H}$: construct the most general representation of $\mathbb{H}$-valued amplitudes on a product state space satisfying C0–C2; show non-commutativity of independent phase updates is unavoidable. For $\mathbb{R}$: formalize the Fourier-dual argument for C5's simultaneous requirements; show $U(1)$ is not intrinsically realizable over $\mathbb{R}$.

T3: Born Rule [Element 5b]

Statement: $P(\psi \to \phi) = |\langle\psi|\phi\rangle|^2$ is the unique countably additive probability measure on Hilbert subspaces.

Upstream: Elements 1, 2a, 2c (T2 needed), 3, 5a; C7 for dimension ≥ 3.

Proof strategy: Gleason's theorem (1957) directly applies given a Hilbert space of dimension ≥ 3. The gap is connecting C7's three spatial dimensions to the state space dimension. This likely requires establishing that the state space carries a representation of the spatial symmetry group (C7 + Wigner's theorem), ensuring state space dimension ≥ 3.

T4: Canonical Commutation Relations [Element 6b]

Statement: $\left[\hat{\phi}(x), \hat{\pi}(y)\right] = i\hbar\delta(x - y)$.

Upstream: Elements 6a, C8, $\hbar$ structural result.

Proof strategy: The Stone-von Neumann theorem guarantees the uniqueness of the irreducible representation of the Heisenberg CCR for finite-dimensional systems. For the infinite-dimensional case, Haag's theorem shows inequivalent representations exist. A representation selection criterion — derivable from C2 (history preservation forbids states not reachable from a natural vacuum) — is needed to close the argument.

T5: Independent Sectors Commute [C9 step]

Statement: Independent equivalence classes under C9 have commuting generators.

Upstream: C1a, C9.

Proof strategy: C1a says independent distinctions continue independently. For Lie group generators, independence of continuation translates to independence of Lie algebra action. Two Lie algebra generators X, Y acting on independent subspaces must commute when those subspaces are C1a-independent. The formal bridge between C1a's categorical statement and the Lie algebra commutator condition needs development.

T6: Gauge Group Uniqueness [C9.3]

Statement: $G = U(1) \times SU(2) \times SU(3)$.

Upstream: T5, quantization bridge, anomaly cancellation, T4.

Proof strategy: This is the longest chain. After establishing the quantization bridge (Elements 1–6), the Ward identities (Section 7.3) give Noether currents. Quantizing these currents yields anomaly conditions. The mathematical requirement that anomalies cancel — enforced by C1 (no continuation can be undefined at the quantum level) — constrains the allowed gauge group and matter representation. The minimum anomaly-free solution with the simplest admissible matter content is $U(1) \times SU(2) \times SU(3)$ with three generations of fermions. This result is known in quantum field theory but has not been derived from C0–C10.

T7: Simply Connected Group Topology

Statement: The admissible topology of G is simply connected.

Upstream: C2, C9.

Proof strategy: C2 forces history to be reachable. A non-simply-connected group admits paths in group space with nontrivial winding number — these are themselves preserved history (winding number is conserved). Whether this forces simple connectivity or permits nontrivial $\pi_1(G)$ is open. Note that $SU(2)$ (simply connected) covers $SO(3)$ (not simply connected) — if C2 forces simple connectivity, this selects $SU(2)$ over $SO(3)$.

T8: Covariant Derivative Uniqueness [C10b]

Statement: $D_\mu = \partial_\mu + iA_\mu$ is the unique local coupling consistent with C10a.

Upstream: C8, C10a.

Proof strategy: Every local modification of ∂_μ consistent with (i) locality, (ii) gauge covariance under C9's equivalence class labels, (iii) linearity (C1+C1a), and (iv) first-order in derivatives (C5 minimality) must be of the form $\partial_\mu + iA_\mu$. Higher-order or non-linear modifications violate one of (i)–(iv). Completeness of this exclusion argument needs formalization.

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11. Full Dependency Graph

The following directed acyclic graph represents the logical structure of the program. Arrows indicate "is required by." A node is available only when all its predecessors are compiled.

```
PRIMITIVES [P]:
C0  C1  C1a  C2  L3  L4  C5
 │   │   │    │   │   │   │
 └───┴───┴────┴───┴───┴───┘
               │
        ┌──────┼──────┐
        ▼      ▼      ▼
    [C]Lin  [C]Lap  [C]C6
        │      │      │
        └──┬───┘      │
           ▼          │
        [C]C8  ←──────┘
           │
    ┌──────┼──────┐
    ▼      ▼      ▼
 [C]C9  [C]ℏ   [T]C7
   │    (struct)  │
   │              ▼
   │           [T]3+1, Huygens
   │
   ▼
 [C]C10a
```

```
     │
     ▼
  [T]C10b
  (covariant derivative)

QUANTUM KINEMATICS:
C1+C1a → [C] Linear state space
C0     → [C] Positive distinguishability
C5     → [C] Dual DOF per amplitude
           ↓
        [T] T2: ℂ unique (Division Algebra Theorem)
           ↓
C2     → [C] Hilbert completion
C0+C5  → [C] Self-adjoint observables
C0+C2+C5 → [C] Probability measure
C7+[C]measure → [T] T3: Born rule (Gleason)
L4+C6+ℏ → [C] Spacelike commutativity
           ↓
        [T] T4: CCR (Stone-von Neumann)

GAUGE STRUCTURE:
[C]C9 + [T]T5 → G is compact Lie, irreducible factors
              → [T] G = U(1)×SU(2)×SU(3) (requires T4 + anomaly bridge)
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12. Conclusions

12.1 What the Framework Establishes

The program has compiled the following results from the primitive set {C0, C1, C1a, C2, L3, L4, C5}:

1. **Linearity** of the redistribution operator (C1+C1a)
2. **Graph Laplacian uniqueness** (five conditions, all derivable)
3. **Ordered history** and the derivation of time as accumulated continuation count (C2+[D])
4. **Causal structure** and the light cone as the shape of the ordering relation (L4+C6)
5. **Free-field wave equation** as the unique extremal of the C8 variational problem
6. **Equivalence class theorem** — internal labels must exist and be preserved (C0+C9)
7. **Gauge principle** — local label symmetry is the unique admissible coupling (C1a+C9+L4)
8. **Compact Lie group structure** for internal symmetry
9. **Linear state space** with positive inner product and Hilbert completion (C1+C1a+C0+C2)
10. **Self-adjoint observables** and the existence of a probability measure (C0+C2+C5)
11. **Spacelike commutativity** and the universal $\hbar$ multiplier (L4+C6+C5)
12. **Space, time, objects, state, and position** all derived — none primitive

12.2 What Remains Candidate

The eight candidate theorems (T1–T8) constitute the open research program:

- **T1** (geometry uniqueness) and **T2** (division algebra) are the most accessible — the proof strategies are well-defined and the mathematical tools exist.
- **T3** (Born rule) and **T4** (CCR) are straightforward applications of known results (Gleason, Stone-von Neumann) once T2 is compiled.
- **T5**, **T6**, **T7**, **T8** form the gauge-group derivation cluster and depend on the quantization bridge. T6 is the deepest — it requires the full anomaly-cancellation argument.

12.3 The Standard of Derivation

The program's standard throughout has been strict: a compiled theorem must have a formal derivation from prior entries with no additional assumptions. Where this standard could not be met, the claim has been demoted to candidate theorem with an explicit proof strategy and upstream dependency.

This discipline has a valuable consequence: the framework is genuinely falsifiable. If T2 (the Division Algebra Theorem) fails — if a valid quantum theory can be built from $\mathbb{R}$-valued amplitudes without violating C5 — then either C5 requires refinement or the argument in §9.3 contains an error. Either way, the framework directs the correction precisely.

If T6 (gauge group uniqueness) fails — if the anomaly cancellation argument permits groups other than $U(1) \times SU(2) \times SU(3)$ — then the Standard Model gauge structure requires an additional primitive. The framework tells you exactly what is missing.

The program's claim is not that it is complete. It is that the architecture is honest: every assumption is visible, every derivation is labeled, and every open problem is precisely stated.

Appendix: Open Problem Summary

Label	Statement	Key dependency	Proof tool
T1	C7 geometry unique	C0–L4	Lattice exclusion argument
T2	$\mathbb{C}$ unique scalar field	C0, C1a, C5	Frobenius + commutativity + phase
T3	Born rule unique	T2, C7 dim ≥ 3	Gleason's theorem
T4	CCR unique	C8, $\hbar$, C6	Stone-von Neumann + rep. selection
T5	Sectors commute	C1a, C9	C1a → Lie algebra commutator
T6	$G = U(1) \times SU(2) \times SU(3)$	T4, T5, anomaly	Anomaly cancellation in QFT
T7	G simply connected	C2, C9	Winding number as C2 history
T8	Covariant derivative unique	C8, C10a	Exhaustion of local couplings

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End of paper.